

Happy Wednesday!

- Quiz 0 is up
- Quiz 1 will release this Friday and due next Monday
- Create your project's team.
 - Use Ed Discussion Megathread to find teammates
 - Due Sept 12th
 - Potential project with Dr. Mick West advertised on Ed
 - Start thinking about your project topic once your team is finalized
- Office hours start this Thursday.
 - In CCB 333
 - Will also occur virtually

Linear Algebra Basics

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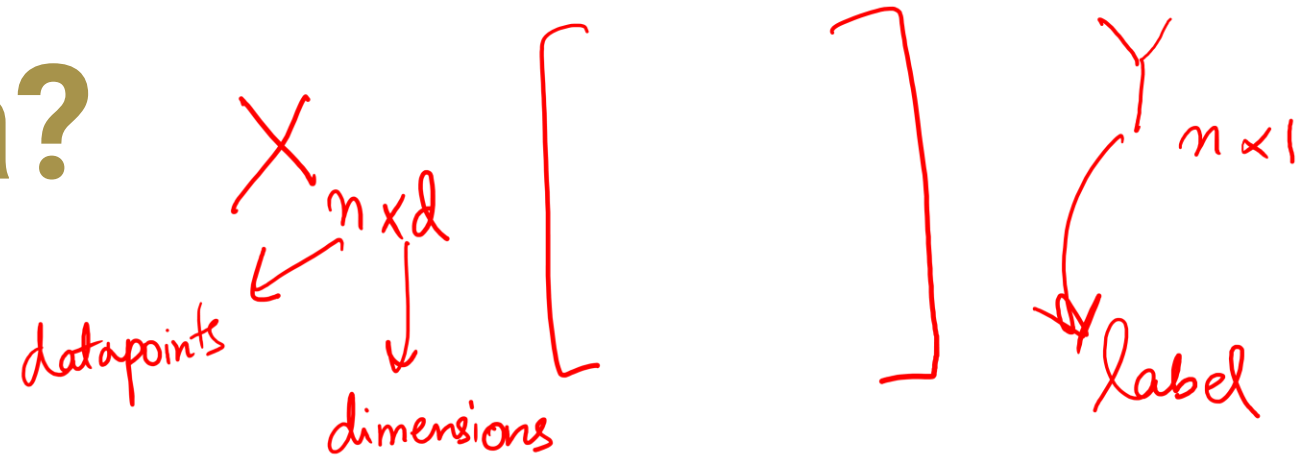
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Slides adapted from Max Roozbahani, Rod Borela, Le Song, Andres Mendez-Vazquez, and Chao Zhang

Outline

- Linear Algebra Basics ←
- Norms
- Multiplications
- Matrix Inversion
- Trace and Determinant
- Eigen Values and Eigen Vectors
- Singular Value Decomposition
- Matrix Calculus

Why linear algebra?



- Most data can be represented or stored in matrix-vector form
- Provides compact representation for sets of linear equations

$$\begin{cases} 4x_1 - 5x_2 = -13 \\ -2x_1 + 3x_2 = 9 \end{cases} \rightarrow \begin{bmatrix} 4 & -5 \\ -2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -13 \\ 9 \end{bmatrix} \rightarrow \mathbf{Ax} = \mathbf{b}$$

- $\mathbf{A} \in \mathbb{R}^{N \times D}$ denotes a matrix with N rows and D columns, where elements belong to real numbers
- $\mathbf{x} \in \mathbb{R}^D$ denotes a vector with D real entries.

$c \in \mathbb{R}$
scalar

Linear algebra basics

- Transpose of a matrix results from flipping the rows and the columns. Given $\mathbf{A} \in \mathbb{R}^{N \times D}$, the transpose is $\mathbf{A}^T \in \mathbb{R}^{D \times N}$

$$\mathbf{A} = \begin{bmatrix} 4 & -5 \\ -2 & 3 \end{bmatrix} \rightarrow \mathbf{A}^T = \begin{bmatrix} 4 & -2 \\ -5 & 3 \end{bmatrix}$$

- For each element of the matrix, the transpose can be written as $A_{ij}^T = A_{ji}$
- The following properties of the transposes are easily verified
 - $(\mathbf{A}^T)^T = \mathbf{A}$
 - $(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$
 - $(\mathbf{A} + \mathbf{B})^T = \mathbf{A}^T + \mathbf{B}^T$

Linear algebra basics

- A square matrix $\mathbf{A} \in \mathbb{R}^{D \times D}$ is symmetric if $\mathbf{A} = \mathbf{A}^T$ and it is skew-symmetric if $\mathbf{A} = -\mathbf{A}^T$. Thus each matrix can be written as a sum of symmetric and anti-symmetric matrices:

$$\mathbf{A} = \underbrace{\frac{1}{2}(\mathbf{A} - \mathbf{A}^T)}_H + \underbrace{\frac{1}{2}(\mathbf{A} + \mathbf{A}^T)}_G$$

$$\begin{aligned} & \frac{1}{2}(\mathbf{A} - \mathbf{A}^T)^T \\ &= \frac{1}{2}(\mathbf{A}^T - \mathbf{A}) \end{aligned}$$

$$\underbrace{\mathbf{A} = \mathbf{A}^T}_{\text{Symmetrical}}$$

$$\underbrace{\mathbf{A} = -\mathbf{A}^T}_{\text{skew-symmetric}}$$

$$H^T = \frac{1}{2}(\mathbf{A}^T - \mathbf{A}) = -\frac{1}{2}(\mathbf{A} - \mathbf{A}^T) = -H \leftarrow \text{skew symmetric}$$

$$G^T = \frac{1}{2}(\mathbf{A} + \mathbf{A}^T)^T = \frac{1}{2}(\mathbf{A}^T + \mathbf{A}) = G \leftarrow \text{symmetric}$$

$$\mathbf{A} = \text{symmetric} + \text{skew symmetric}$$

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Norms

-
- Norm of a vector $\|x\|$ is informally a measure of the length of a vector
 - More formally, a norm is any function $f: \mathbb{R}^D \rightarrow \mathbb{R}$ that satisfies:
 - For all $x \in \mathbb{R}^D$, $f(x) \geq 0$ (non-negativity)
 - $f(x) = 0$ is and only if $x = 0$ (definiteness)
 - For $x \in \mathbb{R}^D$, $t \in \mathbb{R}$, $f(tx) = |t|f(x)$ (homogeneity)
 - For all $x, y \in \mathbb{R}^D$, $f(x + y) \leq f(x) + f(y)$ (triangle inequality)

Norms

ℓ_p

$p = 1, 2, 3, \dots$

$$x = [1, 2, -3]$$

$$\|x\|_2 = \sqrt{1^2 + 2^2 + (-3)^2} = \sqrt{14}$$

- Common norms used in machine learning are:

- ℓ_2 -norm: $\|x\|_2 = \sqrt{\sum_{i=1}^D x_i^2}$

$$\|x\|_2 = \sqrt{x_1^2 + x_2^2 + x_3^2 + \dots}$$

- ℓ_1 -norm: $\|x\|_1 = \sum_{i=1}^D |x_i|$

$$\|x\|_1 = |1| + |2| + |-3| = 6$$

- ℓ_∞ -norm: $\|x\|_\infty = \max_i |x_i|$

$$\|x\|_\infty = \max\{|1|, |2|, |-3|\} = 3$$

manhattan distance

Norms

By default $\rightarrow (\ell_2 \text{ norm})^2$
 $\|x\|_2^2$

- All norms presented so far are examples of the family of ℓ_p norms, which are parametrized by a real number $p \geq 1$

- ℓ_p -norm: $\|x\|_p = \left(\sum_{i=1}^D |x_i|^p\right)^{\frac{1}{p}}$
 $p=1 \rightarrow \ell_1 \text{ norm}$

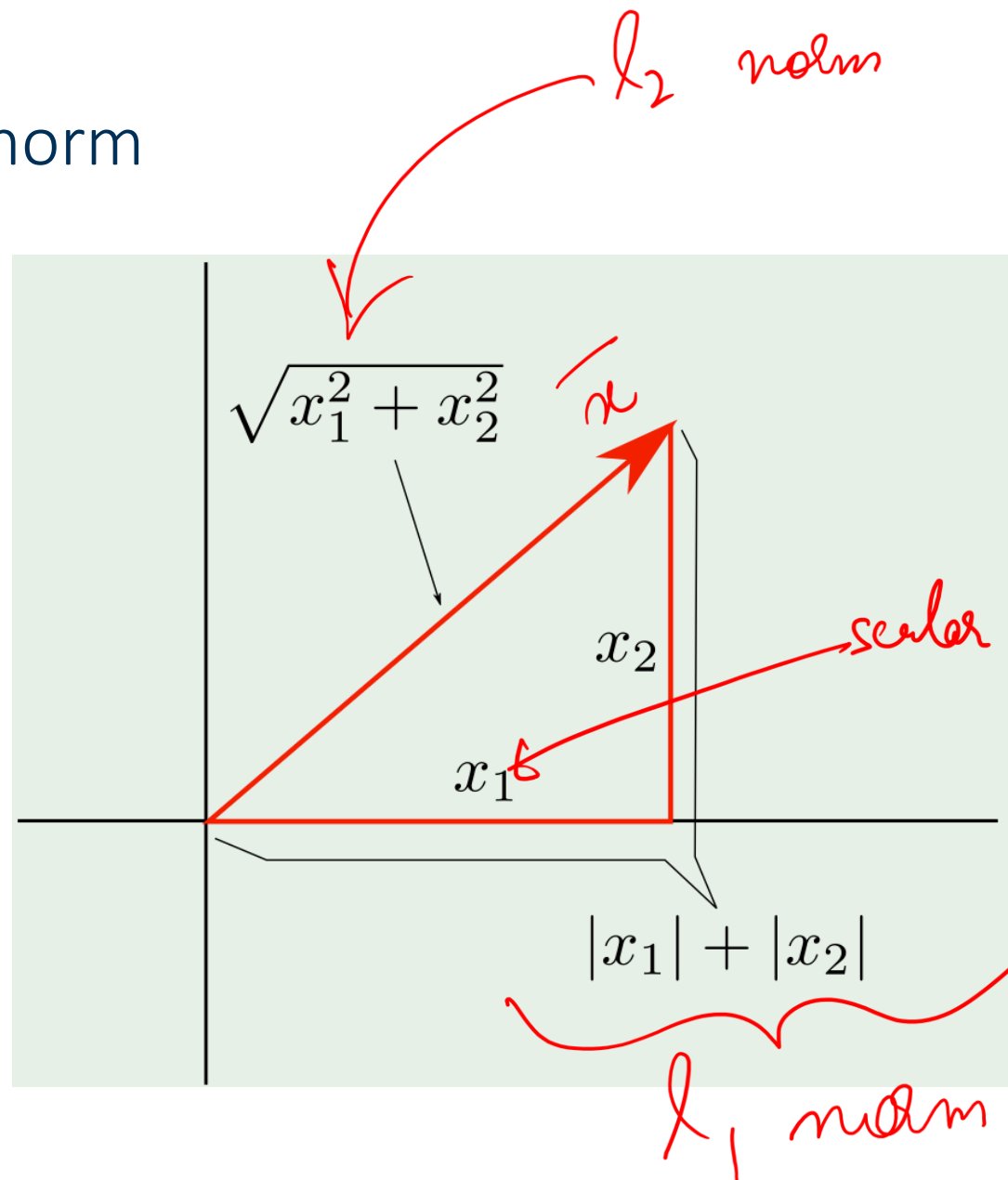
- Norms can be defined for matrices, such as the Frobenius norm

- $\|A\|_F = \sqrt{\sum_{i=1}^N \sum_{j=1}^D A_{ij}^2} = \sqrt{\text{tr}(A^T A)}$

$\| \cdot \|$

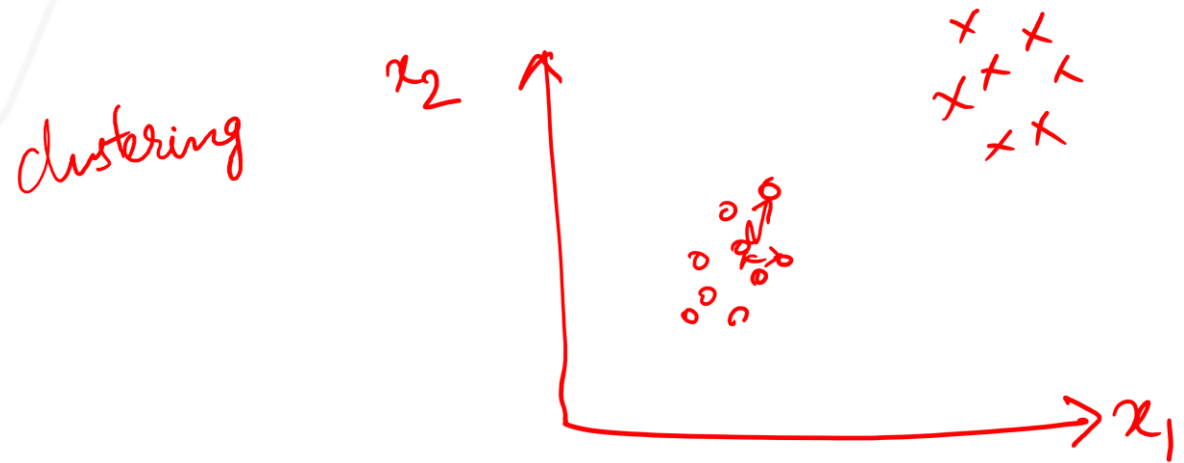
Vector norm examples

- ℓ_1 -norm and ℓ_2 -norm



In context: how are norms useful in ML?

Unsupervised learning

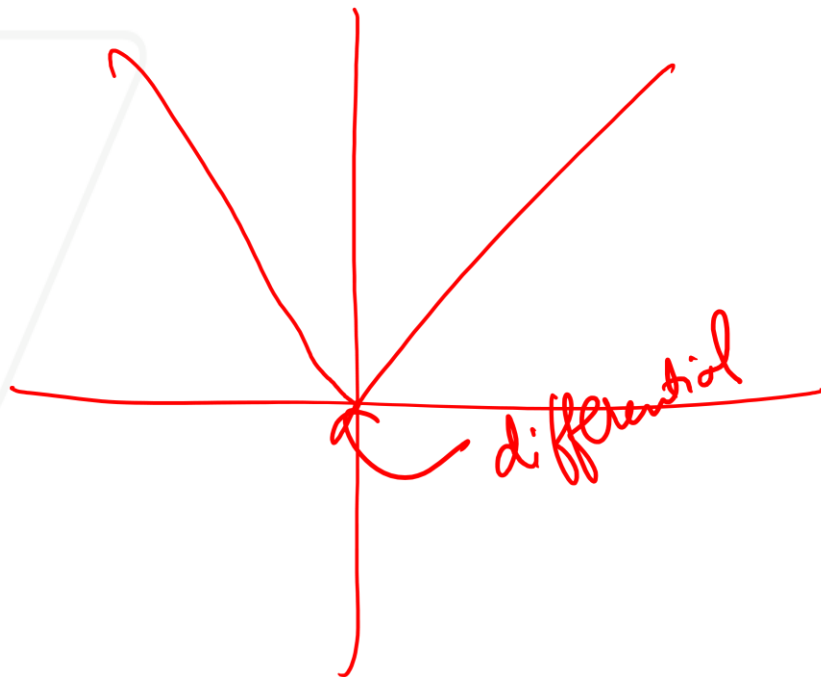


l_2 norm $\rightarrow$ close distances are
in 1 cluster

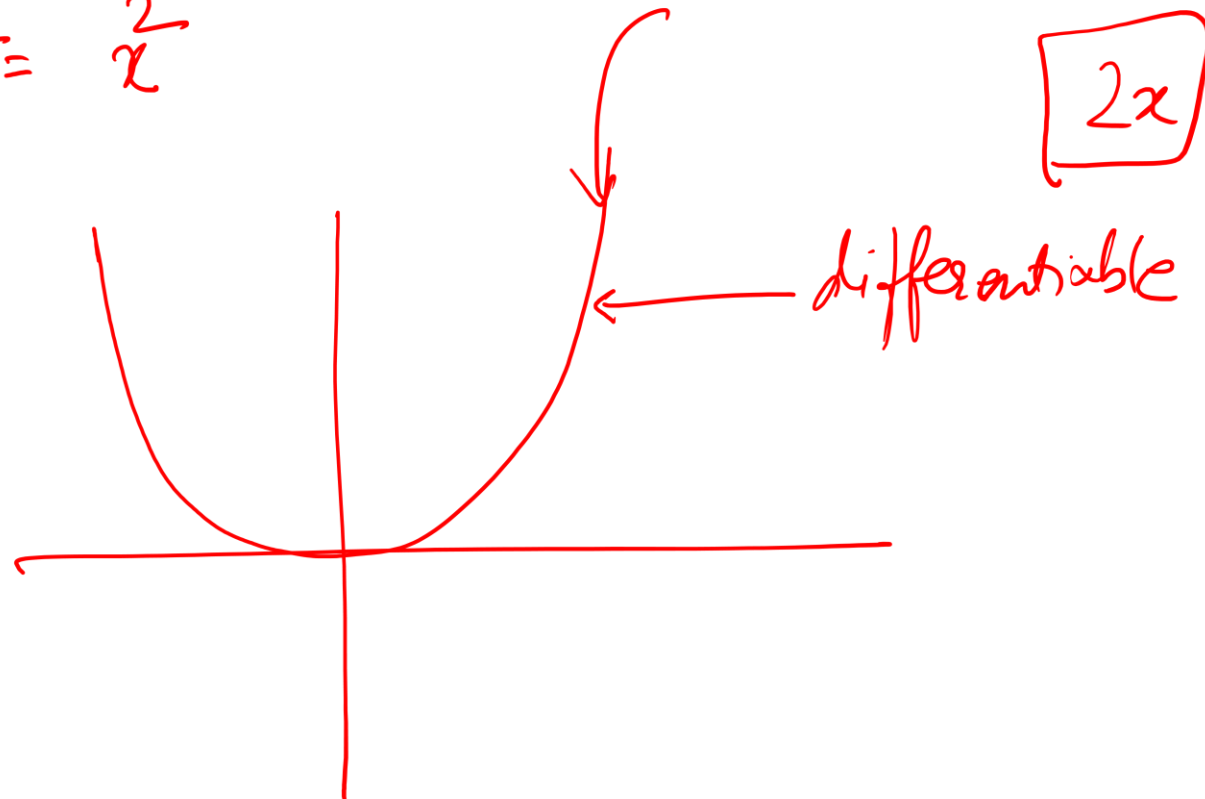
In context: choosing an appropriate norm

Manhattan vs. Euclidean

$$y = |x|$$



$$y = x^2$$



Special Matrices

- The identity matrix, denoted by $I \in \mathbb{R}^{d \times d}$ is a square matrix with ones on the diagonal and zeros everywhere else

$$\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \leftarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

- A diagonal matrix is a matrix where all non-diagonal 'ELEMENTS' are 0. This is typically denoted as $D = \text{diag}(d_1, d_2, \dots, d_d)$

$$\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} = \begin{bmatrix} a^2 & 0 \\ 0 & b^2 \end{bmatrix}$$

Why is diagonal matrix helpful?

- $A^T A =$ elementwise squaring of all diagonal elements
- $A^{-1} = 1/A$ (elementwise)

$$A^{-1} = \begin{bmatrix} 1/a & 0 \\ 0 & 1/b \end{bmatrix}$$

Special Matrices

- Two vectors $x, y \in \mathbb{R}^d$ are orthogonal if $x \cdot y = 0$. A square matrix $U \in \mathbb{R}^{d \times d}$ is **Orthonormal** if all its columns are orthogonal to each other and are normalized

$$x \cdot y = 0$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\leftarrow a \quad \leftarrow b$

$$\frac{a \cdot b = 0}{\|a\|_2 = 1}$$

$$\|b\|_2 = 1$$

unitary $\approx$ orthonormal
 $\leftarrow$ complex values

- It follows from orthogonality and normality that

$$U^T U = I = U U^T$$

$$\|Ux\|_2 = \|x\|_2$$

Is the inverse of a unitary matrix equal to its transpose?

$$U^T U \quad \& \quad U U^T = I$$

$\leftarrow U^T$

$$\begin{bmatrix} U_{d \times d} & x_{d \times 1} \end{bmatrix} \rightarrow \begin{bmatrix} Ux \end{bmatrix}_{d \times 1}$$

$$U U^{-1} = I$$

Orthonormal matrix ONLY rotate a vector. NO STRETCHING or SHRINKING

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Multiplications

Matrix Multiplication
is a bunch of dot products

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \times \begin{bmatrix} a' & b' & c' \\ d' & e' & f' \\ g' & h' & i' \end{bmatrix} = \begin{bmatrix} a a' + b d' + c g' \\ \vdots \end{bmatrix}$$

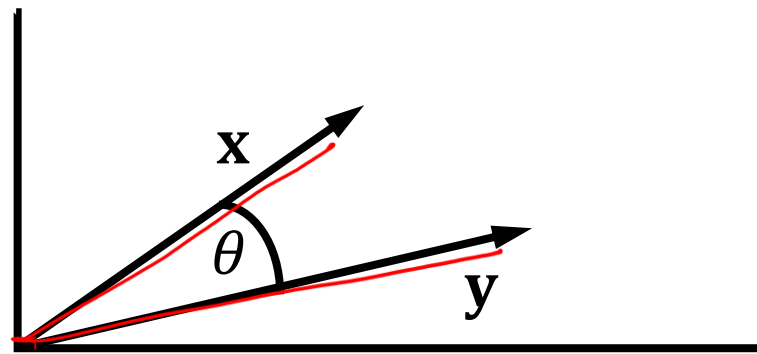
- The product of two matrices $\mathbf{A} \in \mathbb{R}^{N \times D}$ and $\mathbf{B} \in \mathbb{R}^{D \times P}$ is given by $\mathbf{C} \in \mathbb{R}^{N \times P}$, where $C_{ij} = \sum_{k=1}^D A_{ik} B_{kj}$
- Given two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^D$, the term $\mathbf{x} \cdot \mathbf{y}$ (or $\mathbf{x}^T \mathbf{y}$) is called the inner product or dot product of the vectors, and is a real number given by $\sum_{k=1}^D x_i y_i$. For example,

$$\mathbf{x}^T \mathbf{y} = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \sum_{i=1}^3 x_i y_i$$

1x3 3x1 scalar

Multiplications

- The dot product also has a geometrical interpretation, for vectors in $\mathbf{x}, \mathbf{y} \in \mathbb{R}^2$ with an angle θ between them:



$$\mathbf{x} \cdot \mathbf{y} = \|\mathbf{x}\|_2 \|\mathbf{y}\|_2 \cos \theta$$

x project on y

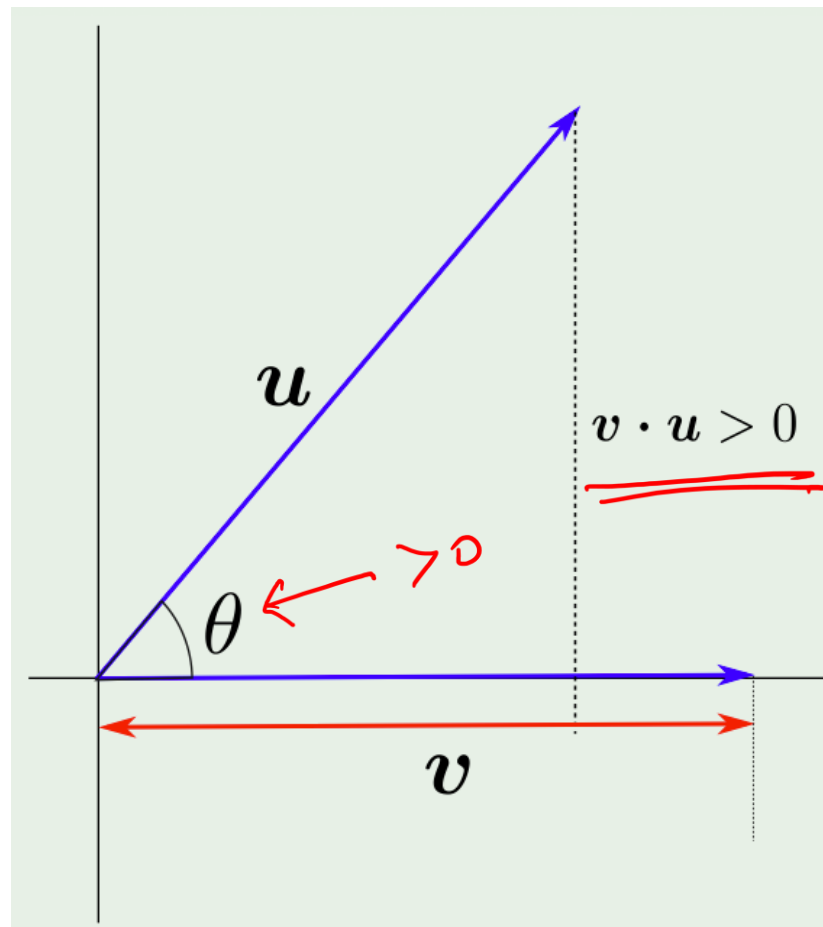
$$= x \cdot \frac{y}{\|y\|}$$

Inner product properties

- The inner product is a measure of correlation between two vectors scaled by the norms of the vectors?

statistical term

loose form



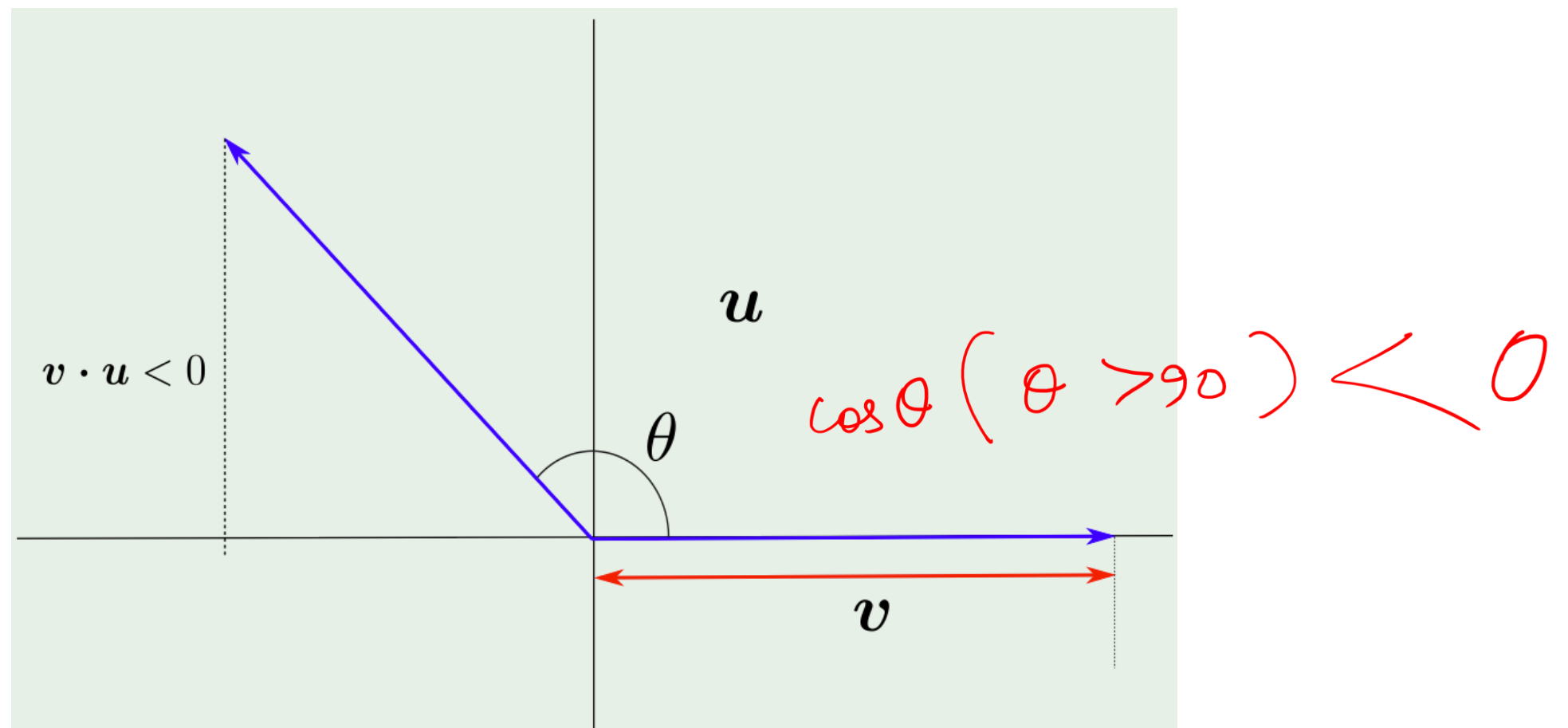
$$\|u\|_2 \cdot \|v\|_2 \cdot \cos \theta$$

$\theta < 90^\circ$
 $\cos \theta > 0$

$$u \cdot v > 0$$

Inner product properties

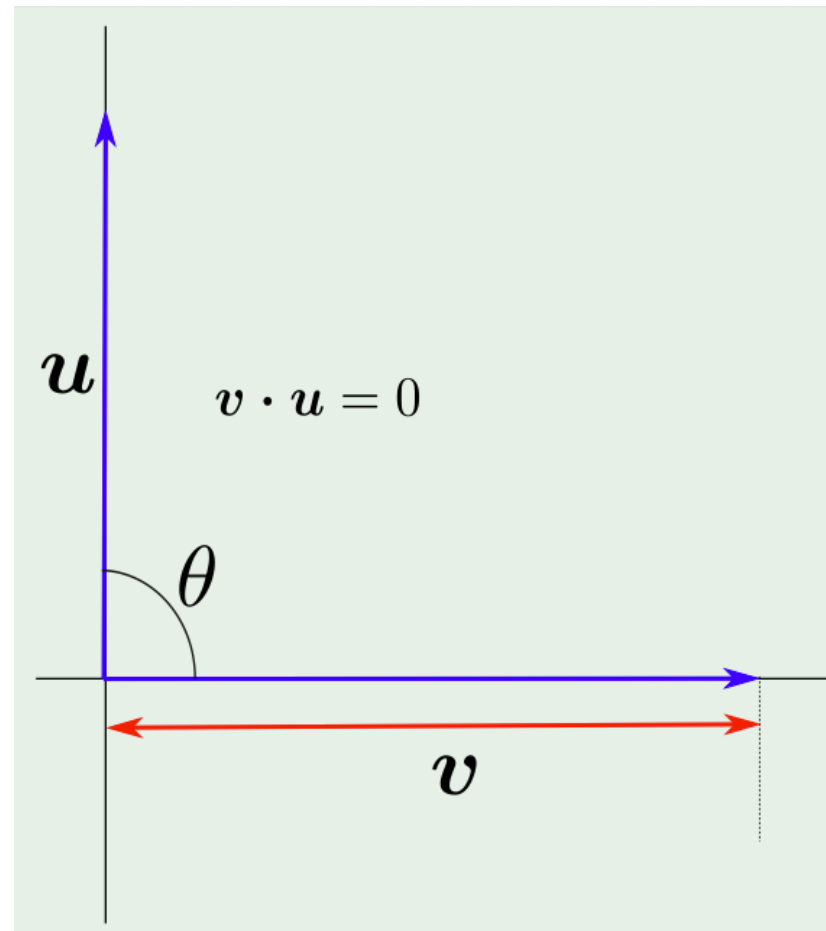
- The inner product is a measure of correlation between two vectors, scaled by the norms of the vectors?



Inner product properties

$a_1 b_1 + a_2 b_2 + a_3 b_3 \rightarrow$ linear operation

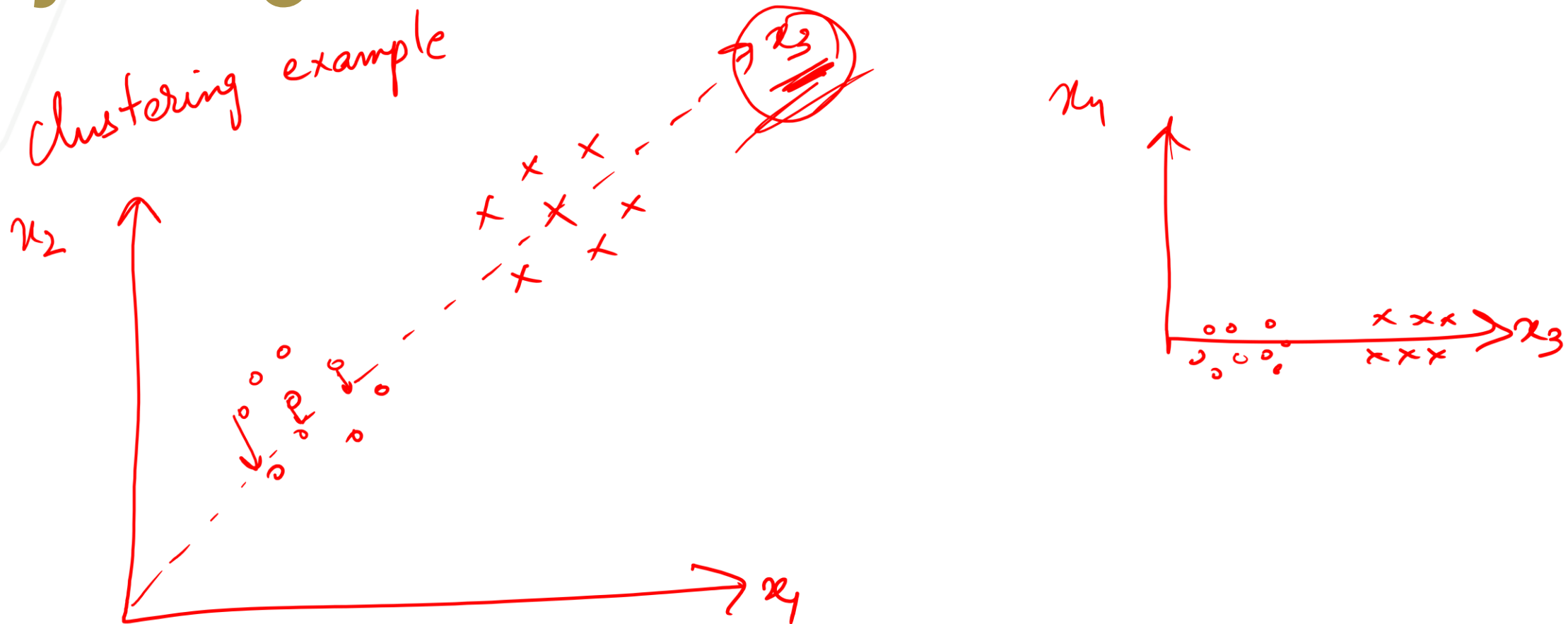
- The inner product is a measure of correlation between two vectors scaled by the norms of the vectors ?



linearly independent
 u & v are

In context: how is the inner product useful in ML?

Projecting data onto a new direction



This will be helpful in dimensionality reduction, classification and feature engineering

Multiplications

- Given two vectors $\mathbf{x} \in \mathbb{R}^N$ and $\mathbf{y} \in \mathbb{R}^D$ the term $\mathbf{xy}^T \in \mathbb{R}^{N \times D}$ is called the outer product of the vectors and is a matrix given by $(x_i y_j)^T = x_i y_j$.
- For example, $\mathbf{x} \in \mathbb{R}^3$ and $\mathbf{y} \in \mathbb{R}^2$

$\mathbf{xy}^T = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} [y_1 \quad y_2] = \begin{bmatrix} x_1 y_1 & x_1 y_2 \\ x_2 y_1 & x_2 y_2 \\ x_3 y_1 & x_3 y_2 \end{bmatrix}$

Handwritten notes:
- $\mathbf{xy}^T \rightarrow \text{scalar}$
- Dimensions: 3×1 (for $\mathbf{x}$) and 1×2 (for $\mathbf{y}^T$)
- Resulting matrix dimension: 3×2
- A red arrow points from the dimensions 3×1 and 1×2 to the resulting 3×2 matrix.

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Linear Independence and Matrix Rank

- A set of vectors $\{x_1, x_2, \dots, x_d\} \subset \mathbb{R}^d$ are said to be **(linearly) independent** if no vector can be represented as a linear combination of the remaining vectors. That is if

$$x_d = \sum_{i=1}^{d-1} \alpha_i x_i$$

$$A = \begin{bmatrix} \vec{a} & \vec{b} & \vec{c} \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$a = \alpha_1 b + \alpha_2 c$
 a, b, c are linearly independent columns or vectors

for some scalar values $\alpha_1, \alpha_2, \dots \in \mathbb{R}$ then we say that the vectors are linearly **dependent**; otherwise the vectors are linearly independent

B is a rank deficient
 $\text{rank}(B) = 2$

$$B = \begin{bmatrix} \vec{a} & \vec{b} & \vec{c} \\ 1 & 4 & 5 \\ 2 & 5 & 7 \\ 3 & 6 & 9 \end{bmatrix}$$

h eye w

$$a + b = c$$

$$c = 1 \times a + 1 \times b$$

linear combination

$$w = h \times 2$$

$A =$ full rank matrix

where all columns are linearly independent

$$\text{rank}(A) = \min(\text{rows}, \text{columns}) = 3$$

In ML, we want as many linearly independent columns as possible

tall matrix $C = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \\ 4 & 8 \end{bmatrix}$ $C_2 = C_1 \times 2$ $\text{rank}(C) = 1$

wide matrix $D = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 10 \end{bmatrix}$ $\text{rank} = 1$
 $\text{rank} = 2$

Linear Independence and Matrix Rank

$N \rightarrow \rightarrow \rightarrow D$

- The **column rank** of a matrix $A \in \mathbb{R}^{n \times d}$ is the size of the largest subset of columns of A that constitute a linearly independent set. **Row rank** of a matrix is defined similarly for rows of a matrix.

It can be easily shown that the row and column ranks are equivalent, therefore we shall refer only to the **rank** of a matrix.

D is a rectangular matrix
 $\min(\text{row}, \text{column}) = 2$

is D full rank $\rightarrow$ Yes

$\text{rank} = 1$

number of rows
 number of columns

In general, for a full rank rectangular matrix, rank is the min of number of rows and number of columns.

Matrix Rank: Examples

What are the ranks for the following matrices? How about an identity matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$$

rank deficient
row 2 = row 1 $\times$ 2
rank (1)

Rank Deficiency means that the number of linearly independent vectors in our set is smaller than the smallest dimension

$$\mathbf{B} = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}$$

Full rank
rank = 3

$x_1 = \alpha_1 x_2 + \alpha_2 x_3$

Matrix Inverse

- The inverse of a square matrix $A \in \mathbb{R}^{d \times d}$ is denoted A^{-1} and is the unique matrix such that $A^{-1}A = I = AA^{-1}$
- For some square matrices A^{-1} may not exist, and we say that A is **singular or non-invertible**. In order for A to have an inverse, A must be **full rank**. *& square $\rightarrow A^{-1}$ exists*
- For non-square matrices the inverse, denoted by A^+ , is given by $A^+ = (A^T A)^{-1} A^T$ called the **pseudo inverse**

In context: why is matrix rank important in ML?

Dataset preprocessing

full rank matrix $\rightarrow X \rightarrow$ all of its features
are not redundant

c

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Matrix Trace

- The trace of a matrix $A \in \mathbb{R}^{d \times d}$, denoted as $\mathbf{tr}(A)$, is the sum of the diagonal elements in the matrix

$$\mathbf{tr}(A) = \sum_{i=1}^d A_{ii}$$

Sum of
diagonal
entries

- The trace has the following properties
 - For $A \in \mathbb{R}^{d \times d}$, $\mathbf{tr}(A) = \mathbf{tr} A^\top$
 - For $A, B \in \mathbb{R}^{d \times d}$, $\mathbf{tr}(A + B) = \mathbf{tr}(A) + \mathbf{tr}(B)$
 - For $A \in \mathbb{R}^{d \times d}$, $t \in \mathbb{R}$, $\mathbf{tr}(tA) = t \cdot \mathbf{tr}(A)$
 - For A, B, C such that ABC is a square matrix $\mathbf{tr}(ABC) = \mathbf{tr}(BCA) = \mathbf{tr}(CAB)$
- The trace of a matrix helps us easily compute norms and eigenvalues of matrices as we will see later

$X = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ v_1, v_2 are dependent

Matrix Determinant

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \det(A) = a(ei - fh) - b(di - fg) + c(dh - eg)$$

Definition (Determinant)

The determinant of a square matrix A , denoted by $|A|$, is defined as

$$\det(A) = \sum_{j=1}^d (-1)^{i+j} a_{ij} M_{ij}$$

where M_{ij} is determinant of matrix A without the row i and column j .

For a 2×2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$$|A| = ad - bc$$



$$X = [v_1, v_2]$$

$X_{3 \times 3}$ = volume of

parallelepiped

Linearly dependent columns $\rightarrow$ matrix is not full rank $\rightarrow$
determinant 0 $\rightarrow$ non-invertible matrix

Properties of Matrix Determinant

Basic Properties

- $|A| = |A^T|$
- $|AB| = |A| |B|$
- $|A| = 0$ if and only if A is not invertible
- If A is invertible, then $|A^{-1}| = \frac{1}{|A|}$.

In context: how is the determinant useful in ML?

Matrix inversion

Testing out if $\det(A) = 0$ before trying to invert a matrix
saves on computation

Recap

$$\begin{bmatrix} d_1 & 0 & 0 \\ 0 & d_2 & 0 \\ 0 & 0 & d_3 \end{bmatrix}$$

- $A^{-1} \rightarrow 1/\text{diagonal elements}$

- $A^T A \rightarrow \text{squaring diagonal elements}$

Orthogonal matrix - $A^{-1} = A^T$
 $\|Ax\|_2 = \text{only rotation } \|x\|_2$

• Special Matrices

• Norms $\mathbb{R}^d \rightarrow \mathbb{R}$ taking the distance of a vector

~~$a \cdot b = 0$~~ unsupervised learning $\rightarrow$ use norms to find distance between datapoints so that we can find clusters

• Matrix Multiplications $\leftarrow$ is a bunch of dot products

$a \cdot b = 0 \Rightarrow$ orthogonal $\Rightarrow$ linearly independent $a, b \neq 0$

• Matrix Inversion $\leftarrow$ A matrix to be invertible $\rightarrow$ linearly independent columns

$\rightarrow$ square matrix

matrix to be rank deficient $\rightarrow$
 we want our data to have as high rank as possible

• Trace and Determinant

$\downarrow$ sum of diagonal elements

$|A| = 0 \rightarrow$ linearly dependent vectors $\rightarrow$ obvious

$\downarrow$ matrix is non-invertible

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Eigenvalues and Eigenvectors

A
 $\begin{bmatrix} & \\ & \end{bmatrix}_{d \times d}$
 d eigenvectors of size $d \times 1$
 $\&$
 d eigenvalues

- Given a square matrix $A \in \mathbb{R}^{d \times d}$ we say that $\lambda \in \mathbb{C}$ is an eigenvalue of A and $x \in \mathbb{C}^d$ is an eigenvector if

$$Ax = \lambda x, \quad x \neq 0$$

matrix A (pointing to A)
 scalar eigenvalue λ (pointing to λ)
 eigenvector x (pointing to x)

$$AX = X\Lambda$$

$d \times d$ (pointing to A)
 $d \times d$ (pointing to X)
 $d \times d$ (pointing to Λ)
 diagonal matrix of d eigenvalues (pointing to Λ)

- Intuitively this means that upon multiplying the matrix A with a vector x , we get the same vector, but scaled by a parameter λ

- Geometrically, we are transforming the matrix A (if symmetric) from its original orthonormal basis/coordinates to a new set of orthonormal basis x with magnitude as λ

$$\begin{bmatrix} A \\ d \times d \end{bmatrix} \begin{bmatrix} \begin{bmatrix} d \times 1 \\ d \times 1 \\ \vdots \\ d \times 1 \end{bmatrix} \\ \begin{bmatrix} d \times d \end{bmatrix} \end{bmatrix} = \begin{bmatrix} X \\ d \times d \end{bmatrix} \begin{bmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \ddots \\ & & & \lambda_d \end{bmatrix}$$

$d \times d$ (pointing to A)
 $d \times 1$ (pointing to each column of the first matrix)
 $d \times d$ (pointing to the second matrix)
 $d \times d$ (pointing to X)
 $\lambda_1, \lambda_2, \dots, \lambda_d$ (pointing to the diagonal elements of Λ)

Matrix Eigen Decomposition

- All the eigenvectors can be written together as $AX = X\Lambda$ where the columns of X are the eigenvectors of A , and Λ is a diagonal matrix whose elements are eigenvalues of A

$$AX \cdot X^{-1} = X\Lambda X^{-1}$$
$$A = X\Lambda X^{-1}$$

- If the eigenvectors of A are invertible, then $A = X\Lambda X^{-1}$
- There are several properties of eigenvalues and eigenvectors

- $Tr(A) = \sum_{i=1}^d \lambda_i$

- $|A| = \prod_{i=1}^d \lambda_i$

$\lambda_2 = 0 \Rightarrow$ matrix is non-inve

- If A is non-singular then $1/\lambda_i$ are the eigenvalues of A^{-1}

- The eigenvalues of a diagonal matrix are the diagonal elements of the matrix itself!

3rd reason
↓
diagonal
matrix

Matrix Eigen Decomposition

- For a symmetric matrix A it can be shown that eigenvalues are real and the eigenvectors are orthonormal. Thus it can be represented as $X\Lambda X^T$

$A \rightarrow$ square
 $A \rightarrow$ Symmetric

$\rightarrow$ eigenvectors become orthonormal
bunch of eigenvector

$$A = X \Lambda X^{-1}$$

$A = X \Lambda X^T$ all eigenvalues in a diagonal matrix

$$\underline{|A - \lambda I| \neq 0} \rightarrow \text{matrix is invertible}$$

Computing Eigenvalues and Eigenvectors

- We can rewrite the original equation in the following manner

$$\begin{aligned} Ax &= \lambda x, & x &\neq 0 \\ \Rightarrow (A - \lambda I)x &= 0, & x &\neq 0 \end{aligned}$$

$$(A - \lambda I)^T (A - \lambda I)x = 0$$

- This is only possible if $(A - \lambda I)$ is singular, that is $\underline{|(A - \lambda I)| = 0}$.

- Thus, eigenvalues and eigenvectors can be computed.
 - Compute the determinant of $A - \lambda I$.
 - This results in a polynomial of degree d .
 - Find the roots of the polynomial by equating it to zero.
 - The d roots are the d eigenvalues of A . They make $A - \lambda I$ singular.
 - For each eigenvalue λ , solve $(A - \lambda I)x$ to find an eigenvector x

Eigenvalue Example

$$|A - \lambda I| = 0$$

$$\left| \begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right| = 0$$

$$\text{Matrix } A = \begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix}$$

$$\left| \begin{array}{cc} 1-\lambda & 2 \\ 3 & -4-\lambda \end{array} \right| = 0 \quad (1-\lambda)(-4-\lambda) - 6 = 0$$

1. Compute the determinant of $A - \lambda I$

$$A - \lambda I = \begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} 1-\lambda & 2 \\ 3 & -4-\lambda \end{bmatrix}$$

$$|A - \lambda I| = (1 - \lambda)(-4 - \lambda) - 6$$

2. Find the roots of the polynomial equating it to zero

$$|A - \lambda I| = 0 \rightarrow (1 - \lambda)(-4 - \lambda) - 6 = 0 \rightarrow \begin{cases} \lambda_1 = -5 \\ \lambda_2 = 2 \end{cases}$$

Eigenvalue Example

matrix $\swarrow$
 A
eigenvector $\searrow$
 x
eigenvalue $\swarrow$
 λ
 $Ax = \lambda x$

x is an eigenvector with eigenvalue $\rightarrow \lambda$

$$\underline{A} \underline{cx} = \underline{c} \underline{Ax} = \underline{c} \lambda x = \lambda (cx)$$

$$A \cdot cx = \lambda (cx)$$

3. For each eigenvalue λ solve $(A - \lambda I)x = 0$ to find eigenvector x

(cx) eigenvector $\rightarrow \lambda$

$$\begin{bmatrix} 1-\lambda & 2 \\ 3 & -4-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow \begin{cases} (1-\lambda)x_1 + 2x_2 = 0 \\ 3x_1 - (4+\lambda)x_2 = 0 \end{cases}$$

Eigenvector for $\lambda_1 = -5$

$$\begin{cases} 6x_1 + 2x_2 = 0 \\ 3x_1 + x_2 = 0 \end{cases} \rightarrow \underline{x_1} = \begin{bmatrix} 1 \\ -3 \end{bmatrix} \text{ or } \begin{bmatrix} 0.3162 \\ -0.9487 \end{bmatrix}$$

Eigenvector for $\lambda_2 = 2$

$$\begin{cases} -x_1 + 2x_2 = 0 \\ 3x_1 - 6x_2 = 0 \end{cases} \rightarrow \underline{x_2} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \text{ or } \begin{bmatrix} 0.8944 \\ 0.4472 \end{bmatrix}$$

Can a matrix have the same eigenvalues?

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \leftarrow \text{same eigenvalues}$$

If two vectors are linearly independent, does it mean they are orthogonal to each other?

No
but
Vice
Versa

$$\begin{bmatrix} 1 & 3 \\ 2 & 5 \end{bmatrix} \leftarrow \begin{array}{l} \checkmark \text{ linearly independent} \\ a \cdot b \neq 0 \end{array}$$

$A \rightarrow$ square & if symmetric $\rightarrow$ eigenvectors are orthogonal

In context: how is eigenvalues and eigenvectors helpful in ML?

- Eigenvectors are “special directions” where a matrix doesn’t twist or mix things — it only stretches or shrinks.
- The stretch amount is the eigenvalue.
- Many ML tools boil down to understanding **data transformations** (via matrices). Eigenvectors/eigenvalues tell us the **essential structure** of those transformations:
- Eigenvectors reveal the fundamental directions along which the transformation acts independently.
- Eigenvalues quantify the strength of that action (how much stretching or shrinking occurs).
- This decomposition simplifies complex transformations into basic “rotate–stretch–rotate back” operations, making it easier to analyze patterns, reduce dimensionality, or understand stability and sensitivity in models.

Outline

- Linear Algebra Basics
- Norms
- Multiplications
- Matrix Inversion
- Trace and Determinant
- Eigen Values and Eigen Vectors
- Singular Value Decomposition



Singular Value Decomposition

$\bar{X}_{n \times d}$ centered $\rightarrow$ along column

n: instances

d: dimensions

X is a centered matrix

$$X = \begin{bmatrix} d_1 & \dots & \dots \\ d_2 & \dots & \dots \\ \vdots & \dots & \dots \\ d_d & \dots & \dots \end{bmatrix}$$

$$\mu_1 = \frac{d_1 + d_2 + \dots + d_d}{d}$$

$$\bar{X} = \begin{bmatrix} d_1 - \mu_1 & \dots & \dots \\ d_2 - \mu_1 & \dots & \dots \\ \vdots & \dots & \dots \\ d_d - \mu_1 & \dots & \dots \end{bmatrix}$$

$$\bar{X} = U \Sigma V^T$$

$U_{n \times n} \rightarrow$ unitary matrix $\rightarrow U \times U^T = I$

$\Sigma_{n \times d} \rightarrow$ diagonal matrix

$V_{d \times d} \rightarrow$ unitary matrix $\rightarrow V \times V^T = I$

singular values

$$X = \begin{bmatrix} u_{1 \times 1} & \dots & \dots & \dots & u_{1 \times n} \\ \vdots & \ddots & \dots & \dots & \vdots \\ \vdots & \vdots & \ddots & \dots & \vdots \\ \vdots & \vdots & \dots & \ddots & \vdots \\ u_{1 \times 1} & \dots & \dots & \dots & u_{n \times n} \end{bmatrix} \times \begin{bmatrix} \Sigma_{1 \times 1} & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \Sigma_{d \times d} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} v_{1 \times 1} & \dots & \dots & \dots & v_{1 \times d} \\ \vdots & \ddots & \dots & \dots & \vdots \\ \vdots & \vdots & \ddots & \dots & \vdots \\ \vdots & \vdots & \dots & \ddots & \vdots \\ v_{d \times 1} & \dots & \dots & \dots & v_{d \times d} \end{bmatrix}$$

$U \qquad \qquad \qquad \Sigma \qquad \qquad \qquad V^T$

$d < n$

Covariance matrix:

eq. 2 *eigenvalue decomposition*

$$C \bar{X} = \bar{X} \Lambda$$

$$C_{d \times d} = \frac{\bar{X}^T \bar{X}}{n}$$

100,000 x 1000

number of datapoints

Do not like list

- ① For Loops (lists)
- ② Inverse
- ③ Matrix Multiplication

$$\bar{X} = U \Sigma V^T$$

$$C = \frac{(U \Sigma V^T)^T (U \Sigma V^T)}{\bar{X} = n U \Sigma V^T}$$

$$= \frac{V \Sigma^T U^T U \Sigma V^T}{n} = \frac{V \Sigma^T \Sigma V^T}{n} = C$$

$$C = \frac{V \Sigma^T \Sigma V^T}{n}$$

$$- CV = \frac{V \Sigma^T \Sigma V^T V}{n}$$

$$C = \frac{\bar{X}^T \bar{X}}{n}$$

eq. 1

$$C = \frac{V \Sigma^T U^T U \Sigma V^T}{n} = \frac{V \Sigma^2 V^T}{n}$$

$$CV = V \left(\frac{\Sigma^2}{n} \right)$$

Singular value decomposition

$$C = \frac{V \Sigma^2 V^T}{n} = V \frac{\Sigma^2}{n} V^T$$

$$CV = V \frac{\Sigma^2}{n} V^T V = V \frac{\Sigma^2}{n}$$

$$CV = V \Lambda$$

Remember:

$$AX = X \Lambda$$

$\lambda_i = \frac{\Sigma_i^2}{n} \rightarrow$ The eigenvalues of covariance matrix

λ_i : Eigenvalue of C or covariance matrix

Σ_i : Singular value of X matrix

So, we can **directly** calculate eigenvalue and eigenvectors of a covariance matrix by having the singular value decomposition of matrix X

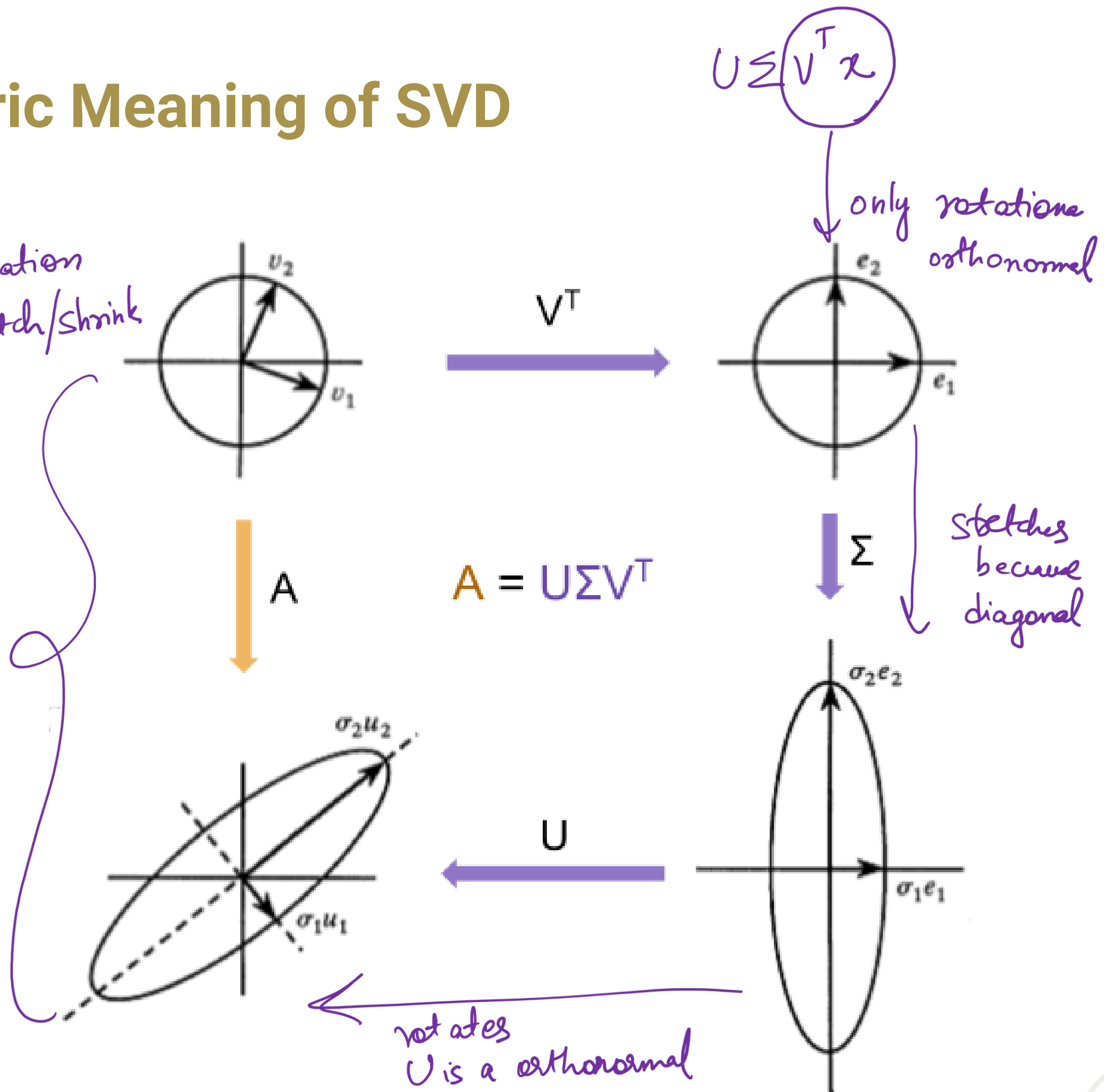
So, we can **directly** calculate eigenvalue and eigenvectors of a covariance matrix by having the singular value decomposition of matrix X

Why is this helpful?

Because SVD **always exists** for **any** matrix (rectangular, non-symmetric) and is **more numerically stable**. That's why in ML (e.g., PCA) we usually prefer **SVD of X** over eigen-decomposition of $X^T X$.

Geometric Meaning of SVD

$Ax =$ rotation
stretch/shrink



In context: why is SVD and covariance matrix useful in ML?

SVD ($X = U\Sigma V^T$)

- Finds the **most informative directions** in the data (columns of V); σ_i tell how strong each is.
- Gives the **best low-rank approximation** → dimensionality reduction, compression, **denoising** (Low values of σ_i means often represents noise)

Covariance ($C = \frac{X^T X}{n}$ with centered X)

- Summarizes **spread** (variances) and **relationships** (correlations) between features.
- Drives **PCA**: eigenvectors = principal components; eigenvalues = **explained variance**.
- Basis for **whitening/standardization**, which helps many models.

→ Covariance tells us what varies; **SVD** shows how to **use** it—rotate to those directions and keep the big values.

Summary

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